

Exact Polynomial Synthesis by Finite Rvachev Up-Function Dictionaries

Dyadic zero multiplicities, cyclotomic nullspaces, Thue–Morse certificates, and output-sensitive algorithms

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Abstract

Let $u = \text{up}$ be Rvachev’s nonnegative C^∞ function supported on $[-1, 1]$ and normalized by $\int u = 1$. This report solves a concrete synthesis problem: represent an arbitrary polynomial exactly on a prescribed compact interval by a finite signed sum of shifted and scaled copies of u .

For a polynomial of degree at most d , the minimal dyadic scale-to-grid ratio is $M = 2^d$. A direct Strang–Fix construction first gives a global cardinal identity with polynomial coefficient sequence $Q = M_u(MD)^{-1}P$, where M_u is the moment generating function of u . Restriction to one unit cell gives $2M$ active atoms. We then determine the complete dyadic alias nullspace. Its characteristic polynomial is

$$A_d(z) = \prod_{j=1}^d \frac{1 - z^{2^j}}{1 - z} = \prod_{\ell=1}^d (z^{2^{\ell-1}} + 1)^{d-\ell+1},$$

with degree $2M - d - 2$. Consequently, the local up-spline space has the unexpectedly small exact dimension $d + 2$. Every polynomial of degree at most d therefore has a unique representation in a proved endpoint basis of only $d + 2$ common-scale atoms. This count is sharp for any fixed dictionary of nontrivial affine up-atoms whose span is required to contain all of Π_d , not merely for the dyadic dictionary.

The recurrence reduction can itself be compressed. An output-sensitive algorithm computes the $d + 2$ coefficients without constructing any exponentially long vector. Its central differential symbol is

$$H_d(z) = \left(\frac{z}{2 \sinh(z/2)} \right)^d \frac{1}{M_u(z)},$$

whose logarithm is an explicit Bernoulli series. For an interval divided into J normalized cells, the same method gives exactly $J + d + 1$ atoms and a scale-localization tradeoff. We also obtain: a one-scalar necessary and sufficient polynomiality certificate; a particularly simple Thue–Morse formula for that certificate; a canonical one-dimensional nonpolynomial defect and its complete Fourier series; rationality and denominator control; exact local dimension and uniqueness statements; affine covariance; numerical conditioning diagnostics; and several conjectures concerning centrally supported bases, arbitrary nonlattice atoms, optimal conditioning, and multivariate extensions.

The basic fact that atomic functions can reproduce polynomials is part of the historical Rvachev literature. The novelty claims made here are deliberately narrower: they concern the explicit finite common-scale construction, the full alias-nullspace and q -integer factorization, the sharp $d + 2$ local dimension/minimality theorem, the output-sensitive Bernoulli algorithm, the Thue–Morse certificate, and the defect formula, relative to both the audited repository and the sources located in the literature search. No claim of global priority is made without qualification.

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1 Problem, answer, and status of the results

1.1 The synthesis problem

Let $p \in \mathbb{R}[x]$ and let $I = [a, b]$ with $a < b$. We seek a finite identity

$$p(x) = \sum_{r=1}^N c_r u\left(\frac{x - \tau_r}{\sigma_r}\right), \quad x \in [a, b], \quad (1)$$

with nonzero scales σ_r , real shifts τ_r , and real amplitudes c_r . Signed amplitudes are essential in general. The requested identity is functional, not merely interpolation at finitely many nodes.

The principal construction in this report uses a common positive scale and an arithmetic grid of shifts. It is already strong enough to solve (1) for every polynomial and interval, with only degree-plus-two atoms when one accepts a broad support footprint.

1.2 The main finite formula

Let $n = \deg p$ and choose an integer $d \geq n$. Choose an integer localization parameter $J \geq 1$, set

$$h = \frac{b - a}{J}, \quad M = 2^d, \quad P(t) = p(a + ht), \quad (2)$$

and compute the rational-linear functionals $b_0(P), \dots, b_{J+d}(P)$ by Algorithm 9.3. Then

$$p(x) = \frac{1}{M} \sum_{r=0}^{J+d} b_r(P) u\left(\frac{x - a - h(M - d - 1 + r)}{Mh}\right), \quad a \leq x \leq b. \quad (3)$$

For $J = 1$, exactly $d + 2$ atoms occur. Increasing J decreases the physical scale $Mh = 2^d(b - a)/J$ but raises the atom count to $J + d + 1$.

Central answer. Every degree- d polynomial on every nondegenerate compact interval is exactly a finite signed sum of $d + 2$ shifted copies of one common dilation of the Rvachev up-function. The shifts and dilation are explicit, all coefficients are obtained by a finite exact algorithm, rational input produces rational output, and $d + 2$ is optimal among all fixed dictionaries of nontrivial affine up-atoms that reproduce the whole space Π_d .

1.3 Status labels and novelty boundary

The report distinguishes four kinds of statements.

- **Imported facts** are taken from the audited repository or standard shift-invariant-space theory: the sinc product, exact integer zero multiplicities, partition of unity, the moment generating function, the Rvachev–Appell inverse, nowhere analyticity, and the general Strang–Fix principle.
- **Proved here** means that a complete mathematical derivation is supplied in this report. These include the orientation with up-atoms as the synthesized functions, the cyclotomic alias nullspace, local dimension, endpoint basis, fast algorithm, certificate, and defect formula.
- **Computational evidence** is exact symbolic experimentation or independent numerical quadrature supplied by the accompanying Python program.
- **Conjectural** statements are explicitly labelled and are not used in proofs.

Polynomial reproduction by shifts of atomic functions is historically known; see Rvachev’s survey and later atomic-function literature [8, 17]. The classical Fourier criterion is the Strang–Fix theory [11, 12]. Thus the existence theorem by itself is not presented as a global first. The finer finite-dimensional and algorithmic package below was not found in the audited ProveIt corpus, and no identical package was located in the targeted literature search.

2 Repository audit and inherited structure

2.1 Scope of the source audit

The source tree under

`Analysis/FabiusFunction/docs`

was recursively inventoried at the 28 August 2026 repository snapshot. The audit included the primary expository documents, the papers directory, the archive and its provenance maps, and every standalone LaTeX draft listed by the reorganized frontier manifest. Because the archive contains superseded and consolidated versions of many reports, the comparison was performed at source/formula level: document inventories and provenance maps were combined with searches for polynomial reproduction, shift synthesis, Strang–Fix conditions, Appell/Kabaya–Iri polynomials, aliasing, root-of-unity filters, and finite up-function sums. The documents most directly relevant to the present problem were then read closely around every matching formula and proof.

The current draft manifest groups the live frontier corpus into Fourier decay, Thue–Morse, exponents and q -series, spectra and arithmetic, integration and transforms, inverse and sampling, representations, and broad frontier compilations [4]. This report is written to complement that organization rather than repeat it.

2.2 Facts imported from the primary synthesis

We use the normalization in the repository’s main synthesis [1]. The Rvachev function $u = \text{up}$ is even, nonnegative, supported on $[-1, 1]$, belongs to $C_c^\infty(\mathbb{R})$, and has integral one. With

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx, \quad (4)$$

its Fourier transform is the entire product

$$\widehat{u}(\xi) = \prod_{j=0}^{\infty} \text{sinc}\left(\frac{\pi \xi}{2^j}\right), \quad \text{sinc } z = \frac{\sin z}{z}. \quad (5)$$

The zero set is exactly $\mathbb{Z} \setminus \{0\}$, and

$$\text{ord}_{\xi=m} \widehat{u} = 1 + v_2(m), \quad m \in \mathbb{Z} \setminus \{0\}. \quad (6)$$

The repository also proves the Poisson partition identities

$$\sum_{k \in \mathbb{Z}} u\left(t + \frac{k}{m}\right) = m \quad (m \in \mathbb{N}), \quad \sum_{k \in \mathbb{Z}} u(t + k) = 1. \quad (7)$$

2.3 Moment and Appell data imported from the inverse/sampling volume

The moment generating function is

$$M_u(z) = \int_{-1}^1 e^{zx} u(x) dx = \prod_{j=0}^{\infty} \frac{\sinh(z/2^{j+1})}{z/2^{j+1}}. \quad (8)$$

Its even cumulants are

$$\log M_u(z) = \sum_{r \geq 1} \kappa_{2r} \frac{z^{2r}}{(2r)!}, \quad \kappa_{2r} = \frac{B_{2r}}{2r(1 - 4^{-r})}. \quad (9)$$

The centered Rvachev–Appell polynomials $\mathcal{A}_n$ are defined by

$$\frac{e^{xz}}{M_u(z)} = \sum_{n \geq 0} \mathcal{A}_n(x) \frac{z^n}{n!}, \quad (10)$$

and satisfy

$$M_u(D)\mathcal{A}_n(x) = x^n, \quad \int_{-1}^1 \mathcal{A}_n(x-y)u(y) dy = x^n. \quad (11)$$

The audited inverse/sampling volume also contains the exact self-sampling quadrature and the finite identity

$$x^n = 2^{-N} \sum_{k \in \mathbb{Z}} \mathcal{A}_n\left(x - \frac{k+\theta}{2^N}\right) u\left(\frac{k+\theta}{2^N}\right), \quad n \leq N. \quad (12)$$

In (12), the translated functions are polynomials and the up-values are scalar weights. Our problem has the opposite orientation: translated and dilated copies of u are the functions, and polynomial data become their amplitudes. This distinction is the main nonduplication boundary.

2.4 Nonpolynomiality on every open interval

The repository’s non-elementarity report proves that the Fabius function agrees with no elementary function, hence with no polynomial, on any subset of $[0, 1]$ having nonempty interior [3]. Since

$$u(x) = F(1 - |x|), \quad |x| \leq 1, \quad (13)$$

no active affine copy of u is polynomial on an open interval contained in the interior of its support. This fact will turn an upper dimension bound into an exact dimension theorem.

3 Normalization and the common-scale dictionary

3.1 Normalized atoms

For a scale ratio $\rho > 0$ and $k \in \mathbb{Z}$, define

$$\psi_{\rho,k}(t) = \frac{1}{\rho} u\left(\frac{t-k}{\rho}\right). \quad (14)$$

Then

$$\widehat{\psi_{\rho,k}}(\xi) = e^{-2\pi i k \xi} \widehat{u}(\rho \xi), \quad (15)$$

$\psi_{\rho,k}$ has support $[k - \rho, k + \rho]$, and its integral is one.

The physical interval problem reduces to the normalized interval by (2). Once

$$P(t) = \sum_r c_r u\left(\frac{t - k_r}{M}\right), \quad 0 \leq t \leq J, \quad (16)$$

then replacing t by $(x - a)/h$ gives the corresponding physical shifts and scale.

3.2 Why the minimal global cardinal ratio is dyadic

Theorem 3.1 (Integer scale and exact reproduction order). *Let $\rho > 0$. The integer translates of $\psi_{\rho,0}$ form a partition of unity,*

$$\sum_{k \in \mathbb{Z}} \psi_{\rho,k}(t) = 1, \quad (17)$$

if and only if ρ is a positive integer. If $\rho \in \mathbb{N}$, the dual-lattice zeros of $\widehat{u}(\rho\xi)$ have minimum multiplicity

$$1 + v_2(\rho), \quad (18)$$

so its Strang–Fix polynomial reproduction order is exactly $v_2(\rho)$. Consequently, among cardinal systems of integer translates with global polynomial reproduction, the smallest scale-to-shift ratio capable of reproducing every polynomial of degree at most d is

$$\boxed{\rho = 2^d}. \quad (19)$$

Proof. The n th Fourier coefficient of the one-periodic sum in (17) is $\widehat{u}(\rho n)$. It is one at $n = 0$. It vanishes for every $n \neq 0$ exactly when ρn is a nonzero integer for every $n \neq 0$; the exact zero set of $\widehat{u}$ makes this equivalent to $\rho \in \mathbb{N}$.

For integer ρ and $n \neq 0$, (6) gives

$$\text{ord}_{\xi=n} \widehat{u}(\rho\xi) = 1 + v_2(\rho n) = 1 + v_2(\rho) + v_2(n).$$

The minimum occurs at odd n . The classical necessary-and-sufficient zero conditions for polynomial reproduction by a compactly supported shift generator then give degree $v_2(\rho)$ [11, 12]. The least integer with $v_2(\rho) \geq d$ is 2^d . $\square$

This minimality statement is about the global cardinal system on the integer shift lattice. Within the dyadic common-scale local family, the exact dimension and basis results in Sections 7–8 show that the same choice is also sharp in the relevant fixed-dictionary sense. No assertion is made here that 2^d is the smallest possible ratio among arbitrary noninteger, nonlattice, or target-adaptive local systems.

Remark 3.2. Odd overscaling does not buy polynomial order. If $\rho = 2^d q$ with q odd, the order remains d , while the support and the number of active atoms increase. This is one reason the rest of the report fixes $M = 2^d$.

4 Global cardinal synthesis and polynomial deconvolution

4.1 A Poisson lemma for polynomial coefficients

Fix $d \geq 0$ and put

$$M = 2^d, \quad \psi_k = \psi_{M,k}. \quad (20)$$

Lemma 4.1 (Polynomial coefficient Poisson collapse). *For every polynomial Q of degree at most d ,*

$$\sum_{k \in \mathbb{Z}} Q(k) \psi_k(t) = \int_{-1}^1 Q(t - My) u(y) dy. \quad (21)$$

Both sides are globally defined; the sum is locally finite.

Proof. For fixed t , apply Poisson summation to

$$f_t(x) = Q(x) \frac{1}{M} u\left(\frac{t-x}{M}\right),$$

which is C_c^∞ . The zero-frequency term is the right side of (21). Multiplication by the degree- d polynomial Q expresses every nonzero Fourier sample $\widehat{f}_t(n)$ as a linear combination of derivatives of $\widehat{u}(M\xi)$ through order d , evaluated at $\xi = n$. At $n \neq 0$, (6) gives zero order

$$1 + v_2(Mn) = d + 1 + v_2(n) > d,$$

so every nonzero Poisson alias vanishes. $\square$

4.2 The deconvolution polynomial

Let $D = d/dt$. Because u is even,

$$\int_{-1}^1 Q(t - My)u(y) dy = M_u(MD)Q(t). \quad (22)$$

The operator is triangular with unit diagonal on polynomials and is therefore invertible.

Theorem 4.2 (Global exact up-atom synthesis). *Let $P \in \Pi_d$ and define*

$$Q = M_u(MD)^{-1}P = \exp\left(-\sum_{r \geq 1} \kappa_{2r} \frac{(MD)^{2r}}{(2r)!}\right) P. \quad (23)$$

The exponential truncates. Then

$$P(t) = \frac{1}{M} \sum_{k \in \mathbb{Z}} Q(k)u\left(\frac{t-k}{M}\right), \quad t \in \mathbb{R}. \quad (24)$$

For $P(t) = t^n$,

$$Q(t) = M^n \mathcal{A}_n(t/M). \quad (25)$$

Proof. Combine Lemma 4.1 with (22) and the definition of Q . Formula (25) follows from the generating function (10) after the dilation $z \mapsto Mz$. $\square$

The first differential terms are

$$Q = P - \frac{M^2}{18}P'' + \frac{31M^4}{16200}P^{(4)} - \frac{2347M^6}{42865200}P^{(6)} + \dots \quad (26)$$

4.3 Finite restriction to an interval

On $[0, J]$, only centers

$$k = -M + 1, -M + 2, \dots, J + M - 1 \quad (27)$$

meet the open interval through the interior of their support. Thus (24) immediately becomes a finite sum with $J + 2M - 1$ active atoms. For $J = 1$ there are $2M = 2^{d+1}$ atoms. The rest of the report compresses this exponential count.

5 A first finite compression by partition pairs

On one unit interval, the partition identity has a local two-term form. For $1 \leq r \leq M$ and $0 \leq t \leq 1$,

$$\psi_{r-M}(t) + \psi_r(t) = \frac{1}{M}. \quad (28)$$

Indeed, $(t - r)/M \in [-1, 0]$ and $u(x) + u(x + 1) = 1$ there.

Let $q_k = Q(k)$ and

$$S = \sum_{j=-M+1}^0 q_j. \quad (29)$$

Then the $2M$ -atom sum compresses to $M + 1$ atoms:

Proposition 5.1 (Pair-compressed representation). *For $0 \leq t \leq 1$,*

$$P(t) = \sum_{r=0}^M A_r \psi_r(t), \quad (30)$$

where

$$\begin{aligned} A_0 &= S, \\ A_r &= q_r - q_{r-M}, \quad 1 \leq r < M, \\ A_M &= S + q_M - q_0. \end{aligned} \quad (31)$$

This representation is simple and can be useful numerically, but $M + 1$ is still exponential in d . It is also nonunique once $M + 1 > d + 2$. The exact alias nullspace reveals the true dimension.

6 The complete dyadic alias nullspace

6.1 Root-of-unity annihilations

Let

$$\omega = e^{2\pi i/M}. \quad (32)$$

For $1 \leq j < M$ and $s \geq 0$, consider the exponential-polynomial coefficient sequence

$$n_k^{(j,s)} = k^s \omega^{jk}. \quad (33)$$

Theorem 6.1 (Full dyadic alias family). *For every $1 \leq j < M$ and every $0 \leq s \leq v_2(j)$,*

$$\sum_{k \in \mathbb{Z}} k^s \omega^{jk} \psi_k(t) = 0, \quad t \in \mathbb{R}. \quad (34)$$

The total number of independent relations is

$$R_d = \sum_{j=1}^{M-1} (v_2(j) + 1) = 2M - d - 2. \quad (35)$$

Proof. The Fourier transform of the twisted polynomial comb $\sum_k k^s \omega^{jk} \delta_k$ is supported, with derivatives through order s , on the coset $j/M + \mathbb{Z}$. At $\xi = j/M + n$ the multiplier $\hat{u}(M\xi)$ is evaluated at $j + Mn$. Since $0 < j < M = 2^d$,

$$v_2(j + Mn) = v_2(j), \quad (36)$$

so the zero order is $1 + v_2(j) > s$. The product distribution is zero.

The sequences (33) are the standard generalized eigensequences for distinct roots ω^j , with multiplicity $v_2(j) + 1$. Their restrictions to any R_d consecutive integers are independent by the confluent Vandermonde theorem. Finally, among $1, \dots, 2^d - 1$, exactly 2^{d-r-1} integers have 2-adic valuation r , giving (35). $\square$

6.2 The cyclotomic annihilator

The relation space is the solution space of one recurrence. Define

$$A_d(z) = \prod_{j=1}^{M-1} (z - \omega^j)^{v_2(j)+1}. \quad (37)$$

Theorem 6.2 (q -integer and cyclotomic factorization). *The alias annihilator has the exact factorizations*

$$A_d(z) = \prod_{\ell=1}^d (z^{2^{\ell-1}} + 1)^{d-\ell+1} \quad (38)$$

$$= \prod_{m=1}^d \frac{1 - z^{2^m}}{1 - z} = \prod_{m=1}^d [2^m]_z. \quad (39)$$

It is monic, has positive integer coefficients, is palindromic, and satisfies

$$\deg A_d = R_d = 2M - d - 2, \quad A_d(1) = 2^{d(d+1)/2}. \quad (40)$$

If

$$A_d(z) = \sum_{\ell=0}^{R_d} a_\ell z^\ell, \quad (41)$$

then a_ℓ counts tuples

$$(r_1, \dots, r_d), \quad 0 \leq r_m < 2^m, \quad \sum_{m=1}^d r_m = \ell. \quad (42)$$

In particular, (a_ℓ) is symmetric, unimodal, and log-concave.

Proof. The roots of $z^{2^{\ell-1}} + 1$ are precisely the M th roots ω^j with $v_2(j) = d - \ell$. Their required multiplicity is $d - \ell + 1$, proving (38). Since $[2^m]_z = \prod_{\ell=1}^m (z^{2^{\ell-1}} + 1)$, multiplicities telescope to (39). The degree and mass follow by summing $2^m - 1$ and multiplying 2^m . The coefficient interpretation follows by expanding the product of uniform blocks. Each block is a finite log-concave sequence and convolution preserves log-concavity. $\square$

Remark 6.3 (A probability law hidden in the annihilator). Let U_m be independent and uniform on $\{0, 1, \dots, 2^m - 1\}$. If

$$L = \sum_{m=1}^d U_m, \quad (43)$$

then

$$\Pr(L = \ell) = \frac{a_\ell}{A_d(1)}, \quad \mathbb{E}L = \frac{R_d}{2}. \quad (44)$$

This finite convolution law is the key to the output-sensitive algorithm.

7 Exact local dimension and sharp common-dictionary sparsity

7.1 The one-cell space

Define

$$V_d = \text{span}\{\psi_k|_{[0,1]} : -M + 1 \leq k \leq M\}. \quad (45)$$

Theorem 7.1 (Exact local dimension). *For every $d \geq 0$,*

$$\boxed{\dim V_d = d + 2.} \quad (46)$$

Moreover,

$$V_d \cap \mathbb{R}[t] = \Pi_d. \quad (47)$$

Proof. There are $2M$ active atoms. By [theorem 6.1](#), their coefficient kernel contains $R_d = 2M - d - 2$ independent vectors. Hence $\dim V_d \leq 2M - R_d = d + 2$. By [theorem 4.2](#), $\Pi_d \subset V_d$, so $\dim V_d \geq d + 1$. At least one active atom, for example ψ_0 , is not polynomial on $(0, 1)$ by the nowhere-analyticity result recalled in [section 2](#). Therefore V_d strictly contains Π_d , giving $\dim V_d \geq d + 2$ and hence equality.

If V_d contained a polynomial outside Π_d , that polynomial together with Π_d would span the $(d + 2)$ -dimensional space V_d . Every atom would then be polynomial on $[0, 1]$, contradicting the same nonpolynomiality fact. Thus [\(47\)](#) holds. $\square$

Corollary 7.2 (The alias relations are complete). *The kernel of the finite synthesis map*

$$T_d : \mathbb{R}^{2M} \rightarrow V_d, \quad (q_k) \mapsto \sum_{k=-M+1}^M q_k \psi_k|_{[0,1]}, \quad (48)$$

has dimension R_d and is exactly the restriction of the recurrence solution space

$$A_d(E)n = 0, \quad (En)_k = n_{k+1}. \quad (49)$$

Proposition 7.3 (Universal fixed-dictionary lower bound). *Let I be a nondegenerate interval, and let*

$$f_r(x) = u\left(\frac{x - \tau_r}{\sigma_r}\right), \quad \sigma_r \neq 0, \quad 1 \leq r \leq N, \quad (50)$$

be affine up-atoms whose restrictions to I are not identically zero. If

$$\Pi_d|_I \subseteq \text{span}\{f_1|_I, \dots, f_N|_I\}, \quad (51)$$

then

$$\boxed{N \geq d + 2.} \quad (52)$$

Proof. Every nonzero affine up-atom is nonpolynomial on I : nonzero overlap with its support contains an open subinterval on which an affine copy of u would otherwise agree with a polynomial, contradicting the nowhere-analyticity fact in [section 2](#). If $N \leq d$, dimension is insufficient. If $N = d + 1$ and the span contains the $(d + 1)$ -dimensional space $\Pi_d|_I$, the two spaces are equal, forcing every $f_r|_I$ to be polynomial. This is impossible. $\square$

Corollary 7.4 (Generic target count in the common dyadic dictionary). *A generic polynomial in Π_d cannot be represented by $d + 1$ or fewer active atoms from the finite common-scale dyadic dictionary.*

Proof. Take the finite union, over all subsets of size at most $d + 1$, of their spans intersected with Π_d . Proposition 7.3 makes every such intersection a proper linear subspace, so the union has Lebesgue measure zero in polynomial coefficient space. $\square$

Warning 7.5. Proposition 7.3 settles optimality for every *fixed* affine up-dictionary that must reproduce the whole polynomial space. It does not settle a target-adaptive nonlinear question: can the shifts and scales be chosen separately for one given polynomial so that a generic target uses only $d + 1$ atoms? That stronger rigidity statement is formulated as Conjecture 18.1.

8 A proved $d + 2$ -atom endpoint basis

8.1 Recurrence elimination

Write

$$K = -M + 1, \quad R = R_d = 2M - d - 2. \quad (53)$$

Then

$$K + R = M - d - 1. \quad (54)$$

Let Q be the deconvolution polynomial in [\(23\)](#), and set $q_k = Q(k)$ on the active range.

Algorithm 8.1 (Full recurrence reduction). Let $A_d(z) = \sum_{\ell=0}^R a_\ell z^\ell$, with $a_R = 1$.

1. Set $n_k = q_k$ for $k = K, K + 1, \dots, K + R - 1$.
2. Extend n by the monic recurrence

$$n_{k+R} = - \sum_{\ell=0}^{R-1} a_\ell n_{k+\ell}. \quad (55)$$

3. Put $b_k = q_k - n_k$ for $K + R \leq k \leq M$.

Theorem 8.2 (Endpoint basis and unique sparse synthesis). *The $d + 2$ functions*

$$\mathcal{B}_d^+ = \{\psi_k|_{[0,1]} : M - d - 1 \leq k \leq M\} \quad (56)$$

form a basis of V_d . For every $P \in \Pi_d$, Algorithm 8.1 gives the unique identity

$$P(t) = \frac{1}{M} \sum_{k=M-d-1}^M b_k u\left(\frac{t-k}{M}\right), \quad 0 \leq t \leq 1. \quad (57)$$

If $P \in \mathbb{Q}[t]$, all b_k are rational.

Proof. The recurrence extension n belongs to the complete nullspace (49), so its synthesis is zero. By construction, $q - n$ vanishes at the first R active centers and is supported at the last $2M - R = d + 2$ centers. Thus every element of V_d lies in the span of (56). Since theorem 7.1 gives dimension $d + 2$, the functions form a basis and the representation is unique. Integer recurrence coefficients preserve rationality. $\square$

Together with Proposition 7.3, theorem 8.2 proves a global fixed-dictionary optimality statement: $d + 2$ is the least possible number of nontrivial affine up-atoms in any fixed dictionary whose span contains all of Π_d on the interval.

Reflection $t \mapsto 1 - t$ gives an equally valid left endpoint basis. The right basis is algebraically canonical but geometrically asymmetric; this matters for conditioning and support overhang.

9 The output-sensitive exact algorithm

9.1 Why the exponential recurrence can be avoided

The full annihilator has degree $R \asymp 2^{d+1}$. Yet the desired output has only $d + 2$ numbers. The key is to compute the residual of $A_d(E)$ on the polynomial deconvolution sequence without expanding A_d .

Let

$$A_1 = A_d(1) = 2^{d(d+1)/2}. \quad (58)$$

For the random variable L in (43), palindromicity gives mean $R/2$, and

$$\mathbb{E}e^{z(L-R/2)} = \prod_{m=1}^d \frac{\sinh(2^{m-1}z)}{2^m \sinh(z/2)}. \quad (59)$$

On the other hand, splitting the product (8) at depth d gives

$$M_u(Mz) = M_u(z) \prod_{r=0}^{d-1} \frac{\sinh(2^r z)}{2^r z}. \quad (60)$$

The ratio simplifies dramatically.

Definition 9.1 (Fast differential symbol). Define

$$H_d(z) = \left(\frac{z}{2 \sinh(z/2)} \right)^d \frac{1}{M_u(z)}. \quad (61)$$

Its logarithm is the explicit Bernoulli series

$$\log H_d(z) = - \sum_{m \geq 1} \frac{B_{2m}}{2m(2m)!} \left(d + \frac{1}{1 - 4^{-m}} \right) z^{2m}. \quad (62)$$

Only terms through degree $\deg P$ can act on P .

Theorem 9.2 (Compressed residual identity). Let $Q = M_u(MD)^{-1}P$, $K = -M + 1$, and

$$g_r = (A_d(E)Q)(K + r). \quad (63)$$

Then, for every integer r ,

$$g_r = A_1 [H_d(D)P] \left(r - \frac{d}{2} \right). \quad (64)$$

Proof. Using $a_\ell/A_1 = \Pr(L = \ell)$,

$$(A_d(E)Q)(x) = A_1 \mathbb{E}Q(x + L).$$

Center L and use $K + R/2 = -d/2$ to obtain

$$g_r = A_1 [M_{L-R/2}(D)M_u(MD)^{-1}P] \left(r - \frac{d}{2} \right).$$

The quotient of (59) by (60) is exactly (61). $\square$

9.2 Triangular recovery of the endpoint coefficients

Let $b_0, \dots, b_{d+1}$ denote the recurrence residual coefficients at centers

$$k_r = M - d - 1 + r. \quad (65)$$

Palindromicity $a_{R-j} = a_j$ gives

$$g_r = \sum_{s=0}^r a_{r-s} b_s, \quad 0 \leq r \leq d+1. \quad (66)$$

Since $a_0 = 1$, this is solved without division.

Algorithm 9.3 (Output-sensitive exact synthesis). Input: $P \in \Pi_d$.

1. Set $M = 2^d$, $A_1 = 2^{d(d+1)/2}$.
2. Compute the truncation of $H_d(z)$ through degree $\deg P$ from (62), and form $G = H_d(D)P$.
3. Compute only $a_0, \dots, a_{d+1}$ of $A_d(z) = \prod_{m=1}^d (1 + z + \dots + z^{2^m-1})$ by truncated sliding-window convolutions.
4. For $0 \leq r \leq d+1$, set

$$g_r = A_1 G(r - d/2). \quad (67)$$

5. Solve

$$b_0 = g_0, \quad b_r = g_r - \sum_{s=0}^{r-1} a_{r-s} b_s. \quad (68)$$

6. Return (57) with $b_{k_r} = b_r$.

Theorem 9.4 (Correctness and arithmetic complexity). *Algorithm 9.3 returns exactly the same coefficients as the full recurrence Algorithm 8.1. With standard quadratic truncated-series arithmetic, it uses*

$$\mathcal{O}(d^2) \quad (69)$$

rational arithmetic operations and stores $\mathcal{O}(d)$ rational numbers. In particular, it is polynomial in the output size rather than exponential in the dyadic scale M .

Proof. The identity (64) computes the same g_r as the full annihilator. Equation (66) follows by applying $A_d(E)$ to the recurrence residual and using the fact that its first R entries vanish. Forward substitution therefore recovers the same endpoint vector. Truncated exponentiation of an even series, differential application, low-coefficient convolution, and the triangular solve each require at most quadratic arithmetic work in d . $\square$

10 Several cells: exact sparsity–localization tradeoff

For $J \geq 1$, define the local space

$$V_{d,J} = \text{span}\{\psi_k|_{[0,J]} : -M+1 \leq k \leq J+M-1\}. \quad (70)$$

Theorem 10.1 (Multi-cell dimension and endpoint basis). *For every $J \geq 1$,*

$$\boxed{\dim V_{d,J} = J + d + 1, \quad V_{d,J} \cap \mathbb{R}[t] = \Pi_d.} \quad (71)$$

The functions with consecutive centers

$$M - d - 1, M - d, \dots, M + J - 1 \quad (72)$$

form a basis. Every $P \in \Pi_d$ has a unique representation in this basis with $J + d + 1$ atoms.

Proof. There are $J + 2M - 1$ active atoms, while the same R_d global alias relations remain independent, so

$$\dim V_{d,J} \leq J + 2M - 1 - R_d = J + d + 1.$$

The $d + 2$ endpoint-basis atoms on the first cell are independent. Add the atoms with centers $M + 1, \dots, M + J - 1$. The atom centered at $M + r$ vanishes on $[0, r]$ and is nonzero immediately to the right of r , while all later added atoms still vanish there. Cell-by-cell restriction gives triangular independence, proving the lower bound. A polynomial in $V_{d,J}$ restricts on the first cell to an element of $V_d \cap \mathbb{R}[t] = \Pi_d$, so it has global degree at most d . Recurrence elimination of the first R_d active coefficients leaves exactly the centers (72). $\square$

The fast algorithm changes only in its output length. For $r = 0, \dots, J + d$, compute

$$g_r = A_1[H_d(D)P](r - d/2), \quad (73)$$

and solve the same triangular convolution (68). Only $a_0, \dots, a_{J+d}$ are required. The arithmetic count is

$$\mathcal{O}(d^2 + d(J + d) + (J + d)^2). \quad (74)$$

Substitution of $t = (x - a)/h$ proves the physical formula (3). The right endpoint basis has total support footprint

$$[a - (d + 1)h, b + (2M - 1)h]. \quad (75)$$

Reflection gives the opposite asymmetry. Thus J is a genuine localization knob: larger J means more atoms but a smaller physical scale and smaller overhang.

11 A one-scalar polynomiality certificate

Let an arbitrary active coefficient vector be written in increasing-center order:

$$q = (q_0, \dots, q_{2M-1}), \quad q_j \text{ multiplies } \psi_{K+j}, \quad K = -M + 1. \quad (76)$$

Define the forward difference $\Delta = E - 1$.

Theorem 11.1 (Exact polynomiality certificate). *Set*

$$\Lambda_d(q) = [A_d(E)\Delta^{d+1}q]_K. \quad (77)$$

Then

$$\sum_{j=0}^{2M-1} q_j \psi_{K+j} \text{ is polynomial on } [0, 1] \iff \Lambda_d(q) = 0. \quad (78)$$

When $q_j = (K + j)^{d+1}$,

$$\Lambda_d(q) = A_d(1)(d+1)!. \quad (79)$$

Proof. Every polynomial synthesis coefficient sequence differs from a degree- d deconvolution polynomial sequence by a null sequence. The former is killed by Δ^{d+1} and the latter by $A_d(E)$, so Λ_d vanishes on the complete preimage of Π_d . That preimage has dimension $R_d + (d+1) = 2M - 1$. The functional is nonzero because (79) follows from $\Delta^{d+1}k^{d+1} = (d+1)!$ and $A_d(E)1 = A_d(1)$. Therefore its kernel is exactly the polynomial preimage. $\square$

11.1 The Thue–Morse simplification

Let

$$\varepsilon_r = (-1)^{s_2(r)}, \quad (80)$$

where $s_2(r)$ is the binary digit sum. The annihilator identity (39) gives

$$\begin{aligned} A_d(z)(z-1)^{d+1} &= (z-1) \prod_{m=1}^d (z^{2^m} - 1) \\ &= (-1)^{d+1} (1-z) \prod_{m=1}^d (1 - z^{2^m}) \\ &= (-1)^{d+1} (1-z) \sum_{r=0}^{M-1} \varepsilon_r z^{2^r}. \end{aligned} \quad (81)$$

Corollary 11.2 (Finite Thue–Morse certificate). *The scalar in [theorem 11.1](#) has the exceptionally simple form*

$$\Lambda_d(q) = (-1)^{d+1} \sum_{r=0}^{M-1} \varepsilon_r (q_{2r} - q_{2r+1}). \quad (82)$$

Thus exact polynomiality is decided by one signed first difference over a finite Thue–Morse block.

Thue–Morse bridge. The same finite Prouhet product that annihilates low polynomial moments is the complete codimension-one obstruction to an arbitrary local up-spline being polynomial. The q -integer alias annihilator and the Thue–Morse difference filter are two factorizations of the same operator.

12 The canonical nonpolynomial defect and its Fourier series

The dimension theorem says that V_d is one dimension larger than Π_d . We now identify that extra direction explicitly.

Set

$$m = d + 1, \quad S_m(t) = \sum_{k \in \mathbb{Z}} k^m \psi_k(t). \quad (83)$$

The sum is locally finite. Its zero-frequency Poisson component is

$$\mathcal{P}_{m,M}(t) = \int_{-1}^1 (t - My)^m u(y) dy. \quad (84)$$

Define the periodic alias residue

$$\mathcal{E}_d(t) = S_m(t) - \mathcal{P}_{m,M}(t). \quad (85)$$

Theorem 12.1 (Explicit periodic residue). *The function $\mathcal{E}_d$ is nonzero, C^∞ , one-periodic, and mean zero. It has the absolutely convergent Fourier series*

$$\mathcal{E}_d(t) = -\frac{(d+1)!}{(2\pi i)^{d+1}} \sum_{\substack{n \in \mathbb{Z} \\ n \text{ odd}}} \frac{\widehat{u}(n/2)}{n^{d+1}} e^{2\pi i n t}. \quad (86)$$

Consequently,

$$\mathcal{E}_d(t + 1/2) = -\mathcal{E}_d(t). \quad (87)$$

For $d + 1$ even it is a cosine series; for $d + 1$ odd it is a sine series.

Proof. The Fourier transform of $\sum_k k^m \delta_k$ is

$$\left(-\frac{1}{2\pi i}\right)^m \sum_{n \in \mathbb{Z}} \delta_n^{(m)}.$$

Multiplication by $\widehat{u}(M\xi)$ leaves at $n = 0$ precisely the polynomial distribution corresponding to (84). At nonzero even n , the zero order $1 + v_2(Mn)$ exceeds m . At odd n , the zero order equals m and only the leading derivative survives. The exact leading jet from (6) and the sinc product is

$$\frac{d^m}{d\xi^m} \widehat{u}(M\xi) \Big|_{\xi=n} = -\frac{m! \widehat{u}(n/2)}{n^m}.$$

This gives (86). Superalgebraic Fourier decay gives absolute convergence with all derivatives. Only odd frequencies occur, proving (87); nonvanishing follows because $\widehat{u}(n/2) \neq 0$ for odd n . $\square$

There is an important subtlety. The periodic residue $\mathcal{E}_d$ by itself is *not* in V_d , because otherwise t^{d+1} would also belong to V_d , contradicting (47). The canonical local defect is instead

$$\mathcal{D}_d(t) = S_m(t) - (\mathcal{P}_{m,M}(t) - t^m) = t^{d+1} + \mathcal{E}_d(t). \quad (88)$$

The subtracted term has degree at most d because the leading term of $\mathcal{P}_{m,M}$ is t^m .

Corollary 12.2 (Canonical direct sum and defect coordinate). *One has*

$$V_d = \Pi_d \oplus \text{span}\{\mathcal{D}_d\}. \quad (89)$$

For an arbitrary coefficient vector q , the coefficient of $\mathcal{D}_d$ in this decomposition is

$$c_{\text{def}}(q) = \frac{\Lambda_d(q)}{A_d(1)(d+1)!}. \quad (90)$$

Proof. The restriction of S_m belongs to V_d , and $\mathcal{P}_{m,M} - t^m$ has degree at most d , so $\mathcal{D}_d \in V_d$. Its certificate equals the certificate of the coefficient sequence k^{d+1} , namely $A_d(1)(d+1)!$ by (79); hence it is not in Π_d . The direct sum follows from (46). Since Λ_d vanishes on the synthesis kernel, it descends to a linear functional on V_d . Applying that functional to $f = P + c\mathcal{D}_d$ annihilates $P \in \Pi_d$ and gives (90). $\square$

13 Arithmetic structure and exactness properties

13.1 Linearity and affine covariance

Every stage of the construction is linear in P . Translation and scaling of the physical interval are handled entirely by (2); no new analytic calculation is needed. Consequently, precomputed coefficient functionals for the monomial basis may be assembled for any polynomial.

13.2 Rationality and denominator control

Write

$$\frac{1}{M_u(z)} = \sum_{n \geq 0} \alpha_n \frac{z^n}{n!}, \quad (91)$$

where

$$\alpha_0 = 1, \quad \alpha_2 = -\frac{1}{9}, \quad \alpha_4 = \frac{31}{675}, \quad \alpha_6 = -\frac{2347}{59535}, \quad \alpha_8 = \frac{1904369}{32531625}. \quad (92)$$

For $P \in \mathbb{Z}[t]$,

$$Q(t) = \sum_{n=0}^d \alpha_n M^n \frac{P^{(n)}(t)}{n!}. \quad (93)$$

Since $P^{(n)}/n! \in \mathbb{Z}[t]$, the denominators of all dense values $Q(k)$ divide the least common multiple of the denominators of $\alpha_0, \dots, \alpha_d$. The recurrence has integer coefficients and introduces no additional denominator. Therefore the same bound applies to the sparse b_r ; the raw-up amplitudes b_r/M add only the dyadic factor M .

13.3 Uniqueness and nonuniqueness

The $2M$ -atom dense representation is highly nonunique after restriction to $[0, 1]$: its coefficient affine space has dimension R_d . The pair-compressed representation is generally still nonunique. The endpoint representation is unique because it uses a basis. Alternative $d + 2$ -atom bases, when they exist, give different unique coordinates and may have very different conditioning.

13.4 Support and endpoint flatness

Every up-atom is C^∞ and flat at the boundary of its compact support. Hence the finite representation extends to a compactly supported C^∞ function outside the target interval, while agreeing exactly with the polynomial inside. This gives an explicit smooth compactly supported extension operator for polynomial data. It is not a polynomial extension globally; all nonpolynomial correction is pushed outside or into the one-dimensional defect direction.

14 Explicit examples

The coefficients below multiply raw copies of u , not the normalized ψ_k .

Example 14.1 (Constant). For $d = 0$, $M = 1$,

$$1 = u(t) + u(t - 1), \quad 0 \leq t \leq 1. \quad (94)$$

Example 14.2 (Linear monomial). For $d = 1$, $M = 2$,

$$t = -\frac{1}{2}u(t/2) + u\left(\frac{t-1}{2}\right) + \frac{1}{2}u\left(\frac{t-2}{2}\right). \quad (95)$$

Example 14.3 (Quadratic monomial). For $d = 2$, $M = 4$,

$$\begin{aligned} t^2 = & \frac{13}{9}u\left(\frac{t-1}{4}\right) - \frac{31}{9}u\left(\frac{t-2}{4}\right) \\ & + \frac{49}{9}u\left(\frac{t-3}{4}\right) + \frac{5}{9}u\left(\frac{t-4}{4}\right). \end{aligned} \quad (96)$$

Example 14.4 (Cubic monomial). For $d = 3$, $M = 8$,

$$\begin{aligned} t^3 = & -14u\left(\frac{t-4}{8}\right) + \frac{136}{3}u\left(\frac{t-5}{8}\right) - \frac{208}{3}u\left(\frac{t-6}{8}\right) \\ & + \frac{280}{3}u\left(\frac{t-7}{8}\right) - \frac{106}{3}u\left(\frac{t-8}{8}\right). \end{aligned} \quad (97)$$

For t^4 , $M = 16$, the six raw-up coefficients are shown in [table 1](#).

Table 1: Endpoint representation of t^4 on $[0, 1]$: coefficient c_k multiplies $u((t - k)/16)$.

k	c_k
11	257824/675
12	-358624/225
13	2010496/675
14	-2528896/675
15	661024/135
16	-2567968/675

15 Numerical experiments and conditioning

15.1 Independent Fourier verification

The accompanying program evaluates

$$u(x) = 2 \int_0^\infty \widehat{u}(\xi) \cos(2\pi x \xi) d\xi \quad (98)$$

by a product truncation and fixed Gauss–Legendre quadrature on each unit frequency lobe. This is independent of the coefficient derivation. The exact symbolic fast algorithm was also compared against the full exponential recurrence.

Table 2: Maximum error on a 17-point grid for the sparse representation of t^d . The Fourier integral was cut at $\xi = 80$ with 48 Gauss nodes per unit lobe.

d	maximum absolute error
0	2.7082×10^{-12}
1	2.7088×10^{-12}
2	1.5028×10^{-11}
3	2.8798×10^{-10}
4	1.6227×10^{-8}
5	5.0834×10^{-6}
6	6.9441×10^{-4}

The exact identities do not deteriorate with d ; the increasing residual is floating-point cancellation. The minimal endpoint basis places sample arguments near a very flat support edge and its coefficients grow rapidly. Therefore exact rational coefficient generation and high-precision evaluation of u are recommended.

15.2 Practical basis choices

There is no universally best representation.

- The $d + 2$ endpoint basis minimizes atom count within the common dictionary and has a triangular exact construction, but can be ill-conditioned.
- The $M + 1$ pair compression has more atoms and sometimes much smaller coefficients, but not uniformly.
- A target-aware basis can be chosen by rank-revealing QR or singular-value analysis of high-precision samples, followed by an exact rational solve in the quotient coordinates.
- The nullspace can be used for ℓ^1 , ℓ^2 , or minimax coefficient optimization without changing the represented polynomial.
- Increasing J reduces the dilation in physical units and may substantially improve locality, at the cost of one atom per added cell.

16 A central-basis conjecture

The endpoint basis is proved but asymmetric. A natural alternative is the consecutive central block

$$\mathcal{C}_d = \{\psi_0, \psi_1, \dots, \psi_{d+1}\}|_{[0,1]}. \quad (99)$$

Reduce these atoms to the endpoint basis using the integer recurrence. Let Δ_d^{cen} be the determinant of the resulting $(d + 2) \times (d + 2)$ integer matrix.

Conjecture 16.1 (Central consecutive basis). *For every $d \geq 0$,*

$$\Delta_d^{\text{cen}} \neq 0. \quad (100)$$

Equivalently, $\mathcal{C}_d$ is a basis of V_d .

Exact computation gives the values in [table 3](#) through $d = 12$.

Nonvanishing does not imply good numerical conditioning. Indeed, some low-degree central coordinates are larger than the endpoint coordinates. The conjecture is algebraic, not a stability assertion. Its determinant sequence appears to have rich divisibility structure and may admit a resultant or cyclotomic interpretation.

17 Further extensions

17.1 Integer scales with an odd factor

Let $\rho = 2^d q$ with q odd. [Theorem 3.1](#) proves that the reproduction degree remains d . The root-of-unity alias argument gives a larger annihilator

$$A_{\rho,d}(z) = \prod_{r=0}^d \frac{z^{\rho/2^r} - 1}{z - 1}, \quad (101)$$

whose degree is

$$2\rho - q - d - 1. \quad (102)$$

Table 3: Exact determinant evidence for Conjecture 16.1.

d	Δ_d^{cen}
0	1
1	1
2	3
3	-1
4	255
5	-6145
6	524287
7	-62914561
8	17716740095
9	-9620726743041
10	11047892835893247
11	-25076042725198921729
12	113484369541461161541631

Indeed, the minimum zero multiplicity on the coset $j/\rho + \mathbb{Z}$ is

$$1 + \min\{v_2(j), d\}. \quad (103)$$

This proves an upper bound $q + d + 1$ for the one-cell local dimension. Exact computation strongly suggests that the bound is attained.

Conjecture 17.1 (Odd overscaling dimension). *For $\rho = 2^d q$ with q odd, the one-cell local space generated by the active integer shifts of $\rho^{-1}u((\cdot - k)/\rho)$ has dimension*

$$q + d + 1. \quad (104)$$

Thus every odd overscaling factor adds exactly $q - 1$ nonpolynomial defect directions while adding no polynomial reproduction order.

17.2 Tensor products

For a box in $\mathbb{R}^s$ and coordinatewise degree bounds $d_1, \dots, d_s$, tensoring the one-dimensional construction gives an immediate exact representation by

$$\prod_{j=1}^s (d_j + 2) \quad (105)$$

products of one-dimensional up-atoms. This is exact for the tensor-product polynomial space. For total degree at most d , the tensor count is far above the polynomial dimension $\binom{d+s}{s}$; a multivariate alias-ideal compression is an attractive frontier problem.

17.3 Smooth polynomial extension operators

Formula (3) defines a linear map from Π_d to $C_c^\infty(\mathbb{R})$ that is the identity on $[a, b]$. Unlike generic extension theorems, it has an explicit finite atomic representation. Questions of minimal support, symmetry, norm growth, and compatibility across adjacent intervals connect this construction to partition-of-unity methods, meshfree trial spaces, and exact boundary patches.

18 Conjectures and research directions

Conjecture 18.1 (Target-adaptive atom-count rigidity). *Fix $d \geq 0$ and a nondegenerate interval I . For Lebesgue-almost every $P \in \Pi_d$, there is no identity on I of the form*

$$P(x) = \sum_{r=1}^N c_r u\left(\frac{x - \tau_r}{\sigma_r}\right), \quad N \leq d + 1, \quad \sigma_r \neq 0, \quad (106)$$

when the amplitudes, shifts, and scales may all depend on P . Classify the exceptional polynomials, if any.

The fixed-dictionary lower bound is Proposition 7.3, and the generic finite-dyadic version is Corollary 7.4. The target-adaptive statement is genuinely nonlinear: an uncountable family of parameter-dependent spans replaces the finite union used in the dyadic proof. Resolving it appears to require a rigidity theorem for cancellations among nonanalytic affine copies of u .

Conjecture 18.2 (Quadratic bit growth for normalized monomials). *For the endpoint coefficients of t^d in lowest terms, the maximum numerator and denominator bit lengths are $\mathcal{O}(d^2)$.*

The fast algorithm makes such a result plausible because $\log A_d(1) = \Theta(d^2)$ and all operator series are Bernoulli-rational, but cancellation and Bernoulli denominators make a proof nontrivial.

Question 18.3 (Condition-optimal basis). *Among all $(d + 2)$ -subsets of the active dyadic dictionary, which basis minimizes a chosen stability functional: the L^∞ coefficient norm for all unit-norm polynomials, the smallest sampled singular value, a Lebesgue-type constant, or support overhang?*

Question 18.4 (Defect special values). *Can the periodic residue (86) be evaluated at dyadic rational arguments in closed form through Fabius values, finite Thue–Morse sums, or q -binomial coefficients? Its odd-frequency support and the values $\hat{u}(n/2)$ suggest a direct bridge to the repository’s half-integer spectral arithmetic.*

Question 18.5 (Central determinant sequence). *Find a product, resultant, or recurrence for Δ_d^{cen} . Determine its prime divisors and possible links to Fermat-type factors visible in the values 255 and 524287.*

Question 18.6 (Positive synthesis). *Characterize polynomials that admit nonnegative amplitudes in an overcomplete up-dictionary. Constants do; sign-changing polynomials do not. For strictly positive polynomials, the problem becomes a finite-dimensional cone membership question after choosing a scale and interval.*

Question 18.7 (Beyond base two). *For base- b atomic functions, replace the 2-adic zero divisor by its base- b analogue. Determine the corresponding cyclotomic annihilator, digit-sequence certificate, and local defect dimension. The expected certificate should involve a generalized Prouhet digit character.*

Question 18.8 (Formal verification). *A compact Lean formalization path is:*

1. polynomial-weighted Poisson summation and the global synthesis theorem;
2. root-of-unity comb derivatives and exact alias multiplicities;
3. the q -integer factorization of A_d ;
4. the local dimension argument using the existing nowhere-analyticity theorem;
5. recurrence elimination and the finite Thue–Morse certificate;
6. formal power-series proof of the H_d cancellation.

Most analytic prerequisites already exist in the repository.

19 Conclusions

The polynomial synthesis problem has a clean exact answer because several structures align:

1. the dyadic sinc product gives zero multiplicity $1 + v_2(m)$;
2. the minimal degree- d dilation is therefore 2^d ;
3. moment deconvolution produces the correct polynomial coefficient sequence;
4. all nonzero dyadic alias cosets generate a large, explicitly factored nullspace;
5. its q -integer annihilator reduces 2^{d+1} active atoms to $d + 2$;
6. the same annihilator is a finite convolution law, which cancels against the Rvachev moment product and yields the output-sensitive operator H_d ;
7. after one additional finite difference, the annihilator becomes a finite Thue–Morse filter, producing a one-scalar exactness certificate;
8. the single local nonpolynomial direction has an explicit odd-frequency Fourier residue.

The resulting construction is exact, finite, affine-covariant, rational over rational input, and algorithmically practical in exact arithmetic. Its main unresolved issues are geometric and nonlinear: how to choose a well-conditioned fixed basis with small support, and whether target-dependent mixed shifts and scales can ever beat the $d + 2$ count for a generic individual polynomial.

A Pseudocode for the fast interval algorithm

Listing 1: Exact polynomial synthesis on an arbitrary interval.

```

INPUT: polynomial p(x), interval [a,b], d >= degree(p), integer J >= 1
OUTPUT: coefficients c_r, shifts tau_r, common scale sigma

h <- (b-a)/J
M <- 2^d
P(t) <- p(a+h*t)
N <- J+d

# H_d(z) from its Bernoulli logarithm, truncated at degree(P)
logH(z) <- - sum_{m>=1} B_{2m}/(2m(2m)!)
              * (d + 1/(1-4^{-m})) z^{(2m)}
H(z) <- exp(logH(z)) mod z^{(degree(P)+1)}
G(t) <- H(D) P(t)

# Low coefficients of A_d(z)=prod_{m=1}^d(1+...+z^{(2^m-1)})
a[0..N] <- truncated sliding-window product
A1 <- 2^{(d(d+1)/2)}

for r=0..N:
  g[r] <- A1 * G(r-d/2)

bcoef[0] <- g[0]
for r=1..N:
  bcoef[r] <- g[r] - sum_{s=0}^{r-1} a[r-s]*bcoef[s]

for r=0..N:
  c[r] <- bcoef[r]/M
  tau[r] <- a + h*(M-d-1+r)

```

```

sigma <- M*h
return c, tau, sigma

IDENTITY: p(x)=sum_{r=0}^N c[r] up((x-tau[r])/sigma), a<=x<=b

```

B Low-order operator data

The first terms of H_d are most compactly stated through (62). Truncated to the degree on which they act:

$$\begin{aligned}
H_0(z) &= 1, \\
H_1(z) &= 1 + \mathcal{O}(z^2), \\
H_2(z) &= 1 - \frac{5}{36}z^2 + \mathcal{O}(z^4), \\
H_3(z) &= 1 - \frac{13}{72}z^2 + \mathcal{O}(z^4), \\
H_4(z) &= 1 - \frac{2}{9}z^2 + \frac{857}{32400}z^4 + \mathcal{O}(z^6).
\end{aligned} \tag{107}$$

The first low annihilator coefficients are

d	$(a_0, a_1, \dots, a_{d+1})$
1	(1, 1, 0)
2	(1, 2, 2, 2)
3	(1, 3, 5, 7, 8)
4	(1, 4, 9, 16, 24, 32)
5	(1, 5, 14, 30, 54, 86, 126)

C Accompanying computational files

The archive contains:

- `rvachev_polynomial_representation.py`: exact symbolic algorithms, the fast and full recurrence constructions, pair compression, both polynomiality certificates, determinant experiments, and independent Fourier evaluation;
- `generate_report_data.py`: reproducible generation of the tables;
- `monomial_sparse_coefficients.csv`: exact endpoint coefficients through degree 8;
- `central_basis_determinants.csv`: exact determinant evidence through degree 12;
- `numerical_residuals.csv`: independent Fourier-quadrature residuals;
- `operator_and_annihilator_table.txt` and `exact_checks.txt`: audit summaries.

The exact self-test compares the output-sensitive and full-recurrence algorithms on one, two, and three cells for two polynomial families through degree 10, verifies the two forms of Λ_d , and checks the normalization (79).

D Repository nonduplication checklist

The following adjacent results were found and deliberately not presented as new:

- exact zero multiplicity $1 + v_2(m)$ for $\widehat{u}$;
- partition of unity and Poisson summation;
- dyadic self-sampling quadrature of degree N ;
- the Rvachev–Appell/Kabaya–Iri inverse moment sequence;
- continuous convolutional deconvolution;
- finite reproduction with translated Appell polynomials weighted by sampled up-values;
- Bell–Bernoulli and Thue–Morse arithmetic already developed elsewhere in the corpus.

The following package appears new relative to the audited repository snapshot:

- exact synthesis with shifted/scaled up-atoms as the functions;
- complete finite dyadic alias nullspace and its q -integer annihilator;
- $\dim V_d = d + 2$ and $V_d \cap \mathbb{R}[t] = \Pi_d$;
- a proved, unique, common-scale $d + 2$ endpoint basis and universal fixed-dictionary minimality;
- the output-sensitive H_d algorithm;
- the $J + d + 1$ multi-cell localization theorem;
- the one-scalar certificate and its finite Thue–Morse form;
- the canonical defect coordinate and Fourier residue;
- the central-basis determinant conjecture and exact evidence.

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